

Vol 9 Chapter 3 — Electron Band Structure

Keeper (author pass — deep math/physics revision)

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Chapter 3 — Electron Band Structure

Level 1 — one sentence

Electrons in periodic crystals have energy bands $E_n(\vec{k})$ instead of discrete atomic levels, with band-filling determining whether the material is a metal (partially filled), semiconductor (small gap), or insulator (large gap), and BST identifies the bands as the substrate's K-type spectrum on the periodic ionic boundary condition.

Level 2 — graduate-physicist precision

3.1 Bands from periodicity

Bloch's theorem (Ch 2) gives wave functions $\psi_{n\vec{k}}(\vec{r})$ with energies $E_n(\vec{k})$ — energy is a function of crystal momentum $\vec{k}$ (restricted to first Brillouin zone), labeled by band index n .

Tight-binding approach: start from atomic orbitals, allow hopping between sites. Gives bands as $E(\vec{k}) = E_0 - 2t \sum_i \cos(\vec{k} \cdot \vec{a}_i)$ for nearest-neighbor hopping with amplitude t . Bandwidth $\sim 4zt$ for coordination number z .

Nearly-free electron approach: start from plane waves, treat periodic potential as perturbation. Gives bands with gaps at Brillouin zone boundaries (Bragg reflection mixes degenerate plane waves).

Both approaches give equivalent qualitative pictures; quantitative numerical methods include DFT, GW, dynamical mean-field theory.

3.2 Metals, semiconductors, insulators

The Fermi level E_F separates filled from empty states at $T = 0$. Position relative to bands distinguishes material types:

- Metal: E_F in a band $\rightarrow$ partial filling $\rightarrow$ free electrons $\rightarrow$ high conductivity. Examples: Cu, Al, Na.
- Semiconductor: E_F in a small gap (~ 0.1 - 3 eV) $\rightarrow$ thermal excitation gives some conduction at finite T . Examples: Si (1.1 eV gap), Ge (0.7 eV), GaAs (1.4 eV).
- Insulator: E_F in a large gap (>3 eV) $\rightarrow$ essentially no conduction. Examples: diamond (5.5 eV), SiO_2 (~ 9 eV).

For semiconductors: doping with electron donors (n-type, e.g., P in Si) or acceptors (p-type, e.g., B in Si) shifts Fermi level — basis of transistors and all modern electronics.

3.3 Effective mass and density of states

Near a band extremum, expand: $E(\vec{k}) \approx E_0 + \hbar^2 |\vec{k}|^2 / (2m^*)$ with m^* the effective mass (can be negative for hole-like bands).

m^* ranges widely: $\sim 0.067m_e$ in GaAs conduction band; $\sim 10m_e$ in narrow-band materials like heavy fermions.

Density of states $g(E)$: number of states per energy per volume. For 3D parabolic band: $g(E) \propto \sqrt{E - E_0}$. Crucial for transport, optical, magnetic properties.

3.4 Fermi surface

At $T = 0$, the Fermi surface is the constant-energy surface $E(\vec{k}) = E_F$ in $\vec{k}$ -space. Determines low-temperature electronic properties: - Free-electron metals: sphere - Cu, Ag, Au: nearly spherical with “necks” touching Brillouin zone faces - Layered materials (graphene, cuprates): 2D-like Fermi surfaces

Quantum oscillations (de Haas-van Alphen, Shubnikov-de Haas) probe Fermi surface geometry experimentally.

3.5 Graphene as model

Graphene (single-layer graphite, 2D honeycomb lattice of carbon): nearly free 2D electron system with massless Dirac dispersion $E(\vec{k}) = \pm \hbar v_F |\vec{k}|$ near K, K' points of Brillouin zone. $v_F \approx c/300$ — relativistic-like dispersion!

Discovered Novoselov-Geim 2004 (Nobel 2010). Enabled the entire 2D materials field.

3.6 BST substrate-mechanism reading

The substrate's K-type structure on a periodic ionic lattice produces the band structure $E_n(\vec{k})$. Specific predictions BST makes for materials (Volume 9 catalog): - Iron pnictide T_c predictions (Task #105) - Topological insulator classification (Task #106) - Quantum Hall plateau corrections (Task #108)

3.7 K-audit anchors

- Vol 5 Ch 1: substrate Hilbert space (parent for band structure)
- Vol 5 Ch 9: spin-statistics (electron Pauli filling)

Level 3 — 5th-grader accessibility

In a single atom, electrons have discrete energy levels. In a solid with billions of atoms, those levels broaden into energy bands — continuous (or nearly so) bands of allowed electron energies. The Fermi level is the “water line”: all states below filled, all above empty (at zero temperature). Position of Fermi level relative to bands determines material type: - Metal: Fermi level inside a band → conducts - Semiconductor: Fermi level in small gap → conducts when warmed or doped - Insulator: Fermi level in big gap → doesn't conduct

This is the foundation of all electronics. Transistors, LEDs, lasers, solar cells — all rely on engineering the band structure with dopants and heterojunctions.

What comes next

Chapter 4 develops superconductivity — including BST predictions for cuprates and B12H32.

Where to look this up

- Ashcroft and Mermin Ch 8-11
- Kittel; Marder, *Condensed Matter Physics*
- BST: tasks #105-108